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Patent Public Search | Text View

United States Patent Application Publication

20250283323

Kind Code

A1

Publication Date

September 11, 2025

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BOTTOM CAP FOR FOAM BUILDING PANELS

Abstract

A bottom cap for a foam building panel includes an elongate rear wall; an elongate front wall spaced from and extending parallel to the elongate rear wall; a bottom wall extending at an obtuse inside angle from the elongate rear wall to the elongate front wall thereby to form an open-topped channel; a first set of spaced pedestals extending into the open-topped channel from a region of the bottom wall that is proximal to the elongate rear wall; and a set of spaced apertures extending from inside the open-topped channel through a region of the bottom wall that is proximal to the elongate front wall. The structure of the bottom cap provides capping for the foam building panel but also some spacing and draining capabilities for enabling moisture that has entered between the bottom cap and the foam building panel to escape.

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Appl. No.:	19/070741
Filed:	March 05, 2025

Related U.S. Application Data

us-provisional-application US 63563345 20240309

Publication Classification

Int. Cl.: E04B1/76 (20060101)

U.S. Cl.:

Background/Summary

CROSS REFERENCE [0001] This application claims priority to U.S. Provisional Patent Application Ser. No. 63/563,345 filed on Mar. 9, 2024 entitled “BOTTOM CAP FOR FOAM BUILDING PANELS,” the content of which is incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates generally to building systems, and more particular to a bottom cap for an exterior cladding panel system for buildings.

BACKGROUND

[0003] The construction of a building or dwelling is regulated by a set of standards, referred to collectively when contemplating construction, materials and the like as the building code. The building code is constantly evolving, and is intended to establish a minimum requirement for material dimensions and structural requirements with the objective of providing safe, comfortable and healthy environments.

[0004] Energy efficiency has received significant attention in the evolution of the building code, with a view to increasing energy efficiency or, at least, reducing energy consumption thereby to, in consequence, reduce the levels of CO₂ in the atmosphere and the costs of operating a building. Newer buildings have been increasingly energy efficient, due to being increasingly insulated. However, with the increases in energy efficiency due to insulation, it has also been recognized that careful attention must be paid to ensuring proper ventilation within the building envelope to avoid condensation.

[0005] An area of particular focus for ventilation is the roof space. In roof space, hot moist air from within the building is subject to cooler ambient temperatures, leading to excessive condensation. Such condensation can lead to undesirable deterioration of the building materials. To avoid this, the building code in many jurisdictions mandates ventilation of the roof space to inhibit such condensation.

[0006] One scheme for ensuring adequate ventilation involves providing the building with an overhanging soffit running around the periphery of the building's roof, there being portions of the soffit that are perforated or that otherwise permit fluid flow along with one or more ridge or gable vents at elevated locations to promote continuous air circulation in the roof space. The soffit is an integral part of the building structure and is typically incorporated into the design of the roof trusses and/or of rafters. In particular, the roof extends outward from the wall a sufficient distance to provide the ventilation area through the soffit required by the building code after the wall cladding and fascia has been applied.

[0007] With the ever-present desire to maximize profits by keeping material to a minimum, the overhang of the soffit is often the minimum allowed by the building code at the time the house was built. In such arrangements, the entire soffit may be perforated to meet the ventilation requirements of the building code.

[0008] As for insulation, with new buildings, insulation can be integrated into walls during construction to ensure the buildings comply with the latest insulation requirements in the building code. However, for existing buildings other approaches are required to ensure the buildings better approach the insulation standards of the latest building code. For example, one approach is to add insulation to the exterior wall structure of an existing building. In particular, one may add sheet insulation to the exterior wall structure, using rectangular panels or sheets made of several inches of expanded polystyrene (EPS) such as Styrofoam™ available from Dow Chemical Corporation. After such insulating panels have been installed, a variety of finishes can be used to clad the

exterior walls.

[0009] Adding EPS panels to the exterior of a building involves arranging the panels to abut the overhang. However, this results in several inches of the perforated soffit, corresponding to the several-inches thickness of the panels and any further cladding applied thereto, being blocked. This accordingly reduces free flow of air into and out of the building. As such, while the insulation has been improved, this consequent reduction in air flow can cause the building code to be contravened for failure to maintain adequate ventilation. Furthermore, it is generally not practical or cost effective to entirely restructure soffits to compensate for the blockages using existing techniques.

[0010] U.S. Pat. No. 10,301,817 to Ackerley, the contents of which are incorporated herein by reference, discloses an exterior insulating system for a building includes at least one insulating panel having a wall-facing side and an outward-facing side, a top and a bottom. The at least one insulating panel having a slope at the top that angles downwardly from the wall-facing side to the outward-facing side. The system includes at least one elongated clip member fastenable along a soffit that is extending outwards from a wall of the building, as well as at least one auxiliary panel having a first edge that is dimensioned to be received by the at least one elongated clip member opposite a main body of the at least one auxiliary panel from a second edge of the at least one auxiliary panel that is connectable to the at least one insulating panel. The main body incorporates structure for permitting airflow through the main body.

[0011] In the system disclosed in the '817 patent, a bottom of the insulating panel facing the foot of the building may be left exposed, though it may be useful to cover the bottom of the insulating panel for aesthetic and functional reasons, such as for preventing animals, insects and weather from reaching the bottom of the insulating panel and potentially breaking it into pieces over time.

However, completely covering the bottom of an insulating panel may prevent any moisture that may arise behind the insulating panel from escaping the system through the bottom, causing a moisture buildup which can lead to problems with freezing and decay.

[0012] It is an object of the present disclosure to obviate or mitigate the above disadvantages.

SUMMARY

[0013] In accordance with an aspect, there is provided a bottom cap for a foam building panel, the bottom cap comprising: an elongate rear wall; an elongate front wall spaced from and extending parallel to the elongate rear wall; a bottom wall extending at an obtuse inside angle from the elongate rear wall to the elongate front wall thereby to form an open-topped channel; a first set of spaced pedestals extending into the open-topped channel from a region of the bottom wall that is proximal to the elongate rear wall; and a set of spaced apertures extending from inside the open-topped channel through a region of the bottom wall that is proximal to the elongate front wall.

[0014] In an example, the obtuse inside angle is from about 95 degrees to about 105 degrees.

[0015] In an example, the obtuse inside angle is from about 97 degrees to about 102 degrees.

[0016] In an example, the obtuse inside angle is 97 degrees.

[0017] In an example, the obtuse inside angle is 102 degrees.

[0018] In an example, the bottom cap further comprises: a second set of spaced pedestals extending into the open-topped channel from a region of the bottom wall that is proximal to the elongate front wall, wherein each of the spaced pedestals in the second set extends to a same height with respect to the elongate rear wall as the spaced pedestals in the first set.

[0019] In an example, the elongate front wall is spaced from the elongate rear wall by about 4 inches.

[0020] In an example, the elongate rear wall extends to about 1 inch from the bottom wall.

[0021] In an example, the elongate front wall extends to about 1.5 inches from the bottom wall.

[0022] In an example, the elongate front wall extends to about 1.85 inches from the bottom wall.

[0023] In accordance with another aspect, there is provided a bottom cap for a foam building panel, the bottom cap comprising: an elongate rear wall; an elongate front wall spaced from and extending parallel to the elongate rear wall; a bottom wall extending at an obtuse inside angle from the

elongate rear wall to the elongate front wall thereby to form an open-topped channel; a set of spaced ledges extending into the open-topped channel from the elongate rear wall; and a set of spaced apertures extending from inside the open-topped channel through a region of the bottom wall that is proximal to the elongate front wall.

[0024] In an example, the obtuse inside angle is from about 95 degrees to about 105 degrees.

[0025] In an example, the obtuse inside angle is from about 97 degrees to about 102 degrees.

[0026] In an example, the obtuse inside angle is 97 degrees.

[0027] In an example, the obtuse inside angle is 102 degrees.

[0028] In an example, the bottom cap further comprises: a set of spaced pedestals extending into the open-topped channel from a region of the bottom wall that is proximal to the elongate front wall, wherein each of the spaced pedestals in the second set extends to a same height with respect to the elongate rear wall as the spaced ledges.

[0029] In an example, the elongate front wall is spaced from the elongate rear wall by about 4 inches.

[0030] In an example, the elongate rear wall extends to about 1 inch from the bottom wall.

[0031] In an example, the elongate front wall extends to about 1.5 inches from the bottom wall.

[0032] In an example, wherein the elongate front wall extends to about 1.85 inches from the bottom wall.

[0033] Other examples are shown and described herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0034] FIG. 1 is a front perspective view of a bottom cap for a foam insulating panel, according to an example;

[0035] FIG. 2 is a cross-sectional endwise elevation view of the bottom cap of FIG. 1;

[0036] FIG. 3 is a top plan view of the bottom cap of FIG. 1;

[0037] FIG. 4 is a front perspective view of the bottom cap of FIG. 1 receiving a foam insulating panel;

[0038] FIG. 5 is a cross-sectional endwise elevation view of an exterior insulating system for providing exterior insulation to a building;

[0039] FIG. 6 is cross-sectional endwise elevation view of a bottom cap for a foam insulating panel, according to an alternative example;

[0040] FIG. 7 is cross-sectional endwise elevation view of a bottom cap for a foam insulating panel, according to another alternative example;

[0041] FIG. 8 is cross-sectional endwise elevation view of a bottom cap for a foam insulating panel, according to another alternative example;

[0042] FIG. 9 is cross-sectional endwise elevation view of a bottom cap for a foam insulating panel, according to another alternative example;

[0043] FIG. 10 is a top plan view of the bottom cap of FIG. 9;

[0044] FIG. 11 is a front perspective view of the bottom cap of FIG. 9 receiving a foam insulating panel;

[0045] FIG. 12 is cross-sectional endwise elevation view of a bottom cap for a foam insulating panel, according to another alternative example;

[0046] FIG. 13 is a top plan view of the bottom cap of FIG. 12;

[0047] FIG. 14 is cross-sectional endwise elevation view of a bottom cap for a foam insulating panel, according to another alternative example;

[0048] FIG. 15 is a top plan view of the bottom cap of FIG. 14;

[0049] FIG. 16 is cross-sectional endwise elevation view of a bottom cap for a foam insulating

panel, according to an alternative example;

[0050] FIG. **17** is cross-sectional endwise elevation view of a bottom cap for a foam insulating panel, according to an alternative example;

[0051] FIG. **18** is cross-sectional endwise elevation view of a bottom cap for a foam insulating panel, according to an alternative example; and

[0052] FIG. **19** is cross-sectional endwise elevation view of a bottom cap for a foam insulating panel, according to an alternative example.

[0053] It will be appreciated that the various components disclosed herein are not necessarily shown in proportion in these figures.

[0054] Other aspects and examples will become apparent upon reading the following description.
DETAILED DESCRIPTION

[0055] FIG. **1** is a front perspective view of a bottom cap **5** for a foam insulating panel (the panel itself not shown in FIG. **1**), according to an example. FIG. **2** is a cross-sectional endwise elevation view of bottom cap **5**, and FIG. **3** is a top plan view of bottom cap **5**.

[0056] In this example, bottom cap **5** includes an elongate rear wall **12** and an elongate front wall **14** spaced from and extending parallel to elongate rear wall **12**. In this example, each of rear wall **12** and front wall **14** is planar and extends in a longitudinal direction between a first end **8** and a second end **10**. Bottom cap **5** also includes a bottom wall **16** extending between rear wall **12** and front wall **14**. In this example, bottom wall **16** also planar and extends in the longitudinal direction between first end **8** and second end **10**, such that rear wall **12**, bottom wall **16**, and front wall **14** together form an open-topped channel.

[0057] A dashed line **7** is included in FIG. **2** not to represent an actual component of bottom cap **5**, but to aid with understanding relative angles of components of bottom cap **5**. Dashed line **7** extends from rear wall **12** to front wall **14**, and in the endwise view at right angles to both walls (since they are parallel to each other).

[0058] In this example, bottom wall **16** extends at an obtuse inside angle **13** (an angle greater than 90 degrees but less than 180 degrees) from rear wall **12** to front wall **14**. Because of this, and because rear wall **12** and front wall **14** extend longitudinally in parallel to one another, bottom wall **16** and front wall **14** form an acute inside angle **15** (an angle greater than 0 but less than 90 degrees). The obtuse inside angle **13** is intended to be such that the channel has a sloped bottom when rear wall **12** is vertically-oriented (such as it is in FIG. **1** and FIG. **2**), with the sloped bottom sloping downwards from rear wall **12** towards front wall **14**. The sloped bottom is for the purpose of coaxing moisture that may enter the open-topped channel to flow away from rear wall **12** and towards front wall **14**.

[0059] It will be appreciated that, due to potential variability in installation of bottom cap **5**, bottom cap **5** itself may not be installed such that rear wall **12** is perfectly vertical. Thus, if the obtuse inside angle **13** is too small, a variance in installation that may cause rear wall **12** to be off-vertical by a few degrees counter-clockwise when bottom cap **5** is installed will, in turn, cause bottom wall **16** to either become level (i.e., horizontal) or to slope towards rear wall **12**. If bottom wall **16** is level then moisture will not be coaxed towards front wall **14**, and if bottom wall **16** is sloped towards rear wall **12**, then moisture will be coaxed towards rear wall **12**. For this reason, it may be useful to ensure obtuse inside angle **13** is greater than the expected installation variability, such that even if rear wall **12** is off-vertical by a few degrees counter-clockwise when bottom cap **5** is installed, bottom wall **16** will still slope towards front wall **14**. Therefore, it may be useful for the obtuse inside angle to be at least about 95 degrees (i.e. a downward slope of 5 degrees from a vertical rear wall **12**), providing for approximately a 4 degree variation counter-clockwise from rear wall **12** being vertical, while still providing a frontward-sloped bottom wall **16**. It may alternatively be useful for the obtuse inside angle to be about 97 degree (i.e. a downward slope of 7 degrees from a vertical rear wall **12**), providing for about a 6 degree variation counter-clockwise from rear wall **12** being vertical, while still providing a frontward-sloped bottom wall **16**.

[0060] Furthermore, it will be appreciated that due to manufacturing variability, a given bottom cap 5 may have a slightly-different obtuse inside angle 13 after manufacturing than that of another bottom cap 5. It would be useful to ensure that manufacturing tolerances as well as installation tolerances be borne in mind when selecting the obtuse inside angle 13.

[0061] Still further, it will be appreciated that, depending on the malleability of the material used for bottom cap 5—such as if bottom cap 5 is made from sheet metal—an installer may find that during shipment or handling the angles between rear wall 12, front wall 14, and bottom wall 16 have changed since manufacturing such that they need adjustment at the time of installation. It will also be appreciated that a given installer may decide to create a bottom cap 5 from sheet metal anew, onsite during installation. Such an installer may wish to use a jig to help ensure that the angles are suitable for coaxing any moisture that eventually gets inside of the channel of bottom cap 5 towards front wall 14.

[0062] While the above description about obtuse inside angle 13 explains the practical utility of having obtuse inside angle 13 greater than a certain amount, it will similarly be appreciated that providing an obtuse inside angle 13 that is too great may be impractical. For example, if the obtuse inside angle 13 is close to 180 degrees, then although the slope downwards towards front wall 14 will be assured this will, in turn likely cause bottom wall 16 to be impractically long and for front wall 14 to also be impractically tall to ensure it can aid, along with rear wall 12, with receiving a bottom of a foam building panel into the open-topped channel. Furthermore, the overall height of such a bottom cap 5 would be quite large if the obtuse inside angle 13 were very large, leading to possible installation problems and possible aesthetic problems. For these reasons, it may be useful to ensure obtuse inside angle 13 is greater than an amount such as one of the amounts set forth herein, while also being less than an amount. In particular, it may be useful for the obtuse inside angle to be at least about 95 degrees (i.e. a downward slope of 5 degrees from a vertical rear wall 12) but no greater than 105 degrees (i.e. a downward slope of 15 degrees from a vertical rear wall 12). It may alternatively be useful for obtuse inside angle 13 to be at least about 97 degrees (i.e. a downward slope of 7 degrees from a vertical rear wall 12) but no greater than 102 degrees (i.e. a downward slope of 12 degrees from a vertical rear wall 12). It will be appreciated that other obtuse inside angles 13 may be implemented as desired or required for a given installation, that take into account the considerations set forth herein and other considerations as may arise in a manufacturing, shipping, or installation context.

[0063] With reference particularly to FIG. 2 and FIG. 3, the open-topped channel has a width W for receiving the bottom of a foam insulating panel. In this example, width W is about 4 inches, so as to receive and “cap” the bottom of a foam insulating panel that is about 4 inches thick.

Furthermore, in this example, obtuse inside angle 13 is 97 degrees and acute inside angle 15 is accordingly 83 degrees. In this example, the rear wall 12, front wall 14 and bottom wall 16 are formed from a single piece of sheet metal, bent into position using a roll forming process or another process suitable for bending the single sheet to form walls 12, 14, 16. Because, in this example, the tops of both rear wall 12 and front wall 14 come to the same height when rear wall 12 and front wall 14 are both perfectly vertical, if rear wall 12 is 1 inch tall, then with the 4 inch width W, front wall 14 would be about 1.85 inches tall.

[0064] In this example, bottom cap 5 includes a first set of spaced pedestals 18 extending into the open-topped channel from a region of bottom wall 16 that is proximal to rear wall 12. The spaced pedestals 18 together supports a rear end of the bottom of a foam insulating panel when received within the open-topped channel so that the foam insulating panel does not rest directly on bottom wall 16. In this way, moisture that might come into the channel from the rear while the foam insulating panel is received within the channel is generally able to flow along bottom wall 16, underneath the foam insulating panel and through the spaces between pedestals 18, towards front wall 14. Each one of pedestals 18 may be formed by a punch process that forms indentations into sheet metal from a bottom-facing side of bottom wall 16, thus forming pedestals that extend up into

the open-topped channel. Pedestals **18** may be formed by other processes.

[0065] Furthermore, bottom cap **5** includes a set of spaced apertures **20** extending from inside the open-topped channel through a region of bottom wall **16** that is proximal to front wall **14**, to the exterior of bottom cap **5**. The spaced apertures **20** provide a point of drainage of any moisture within the channel that flows towards front wall **14**, thereby to enable the moisture to escape from the channel. In this example, the spaced apertures **20** extend along the bottommost point in the sloped bottom wall **16**, so that moisture reaching the bottommost point can escape.

[0066] In alternative examples, spaced apertures may instead extend along a point that is slightly farther up the slope, or even very slightly up front wall **14** from the bottommost point and may still drain substantial moisture before or after the moisture reaches the bottommost point. That is, if moisture builds up at the bottommost point it will back up and drain through spaced apertures even if they are not themselves coincident with the bottommost point. In alternative examples, multiple rows or an array of spaced apertures may be provided instead of or in addition to a row of spaced apertures **20**, as long as the integrity of the overall structure is not unduly compromised by the number and frequency of the spaced apertures.

[0067] It will be appreciated that, if the width *W* between rear wall **12** and front wall **14** is similar to the thickness of the bottom of the foam insulating panel, then the foam insulating panel will not easily tend to seat flat against the sloped bottom wall **16** near to front wall **14**. Therefore, moisture in the channel is generally able to flow uninhibited underneath the foam insulating panel, to reach the spaced apertures **20**, and thereby escape the channel. Each one of apertures **20** may be formed by a punch process that forms apertures through bottom wall **16**. Apertures **20** may be formed by other processes.

[0068] It will be appreciated that one or both of the set of pedestals **18** and the set of apertures **20** may, during manufacture/creation, be formed in a length of sheet metal prior to the sheet metal itself being formed into rear wall **12**, front wall **14**, and bottom wall **16**. Alternatively, one or both of the set of pedestals **18** and the set of apertures **20** may, during manufacture/creation, be formed in a length of sheet metal after the sheet metal is formed into rear wall **12**, front wall **14**, and bottom wall **16**. Alternative orders of such steps to ensure quality, ease of manufacture, and cost effectiveness may be used.

[0069] FIG. **4** is a front perspective view of bottom cap **5** receiving a foam insulating panel *P* having a thickness of about 4 inches, corresponding to a width *W* of about 4 inches. It can be seen that, when seated within bottom cap **5**, foam insulating panel *P* rests on pedestals **18** (only one of which is visible in FIG. **4**) and closely between both rear wall **12** and front wall **14**, thus providing a generally wedge-shaped volume formed mainly by the bottom of foam insulating panel *P*, a portion of front wall **14**, and bottom wall **16**, in which moisture can flow generally freely underneath foam insulating panel *P* towards apertures **20** (only one of which is visible in FIG. **4**) so that the moisture can escape to outside of the channel.

[0070] FIG. **5** is a cross-sectional endwise elevation view of an exterior insulating system **100** for providing exterior insulation to a building. System **100** is adapted for use in applying additional insulation to an existing building, but may be used while constructing buildings anew, as the builder sees fit. System **100** includes an insulating panel *P*, in this example formed from expanded polystyrene (EPS) and having a wall-facing side and an outward-facing side. Insulating panel *P* has a top and a bottom. At the top, which is adjacent to a soffit **90** of the building *B* and particularly a soffit channel supporting soffit **90** in its position, there is a slope *S* downward from the wall-facing side to the outward-facing side. In this example, slope *S* is a 45 degree bevel. In FIG. **1**, only one insulating panel *P* is shown, though installations will typically involve several insulating panels *P*. Each insulating panel *P* is fastened to the wall with fasteners *F*.

[0071] System **100** also includes an auxiliary panel **40** connected between soffit **90** and insulating panel *P*, as described in more detail in the '817 patent to Ackerley. In FIG. **5**, one auxiliary panel **40** is shown, though other installations may involve more than one auxiliary panel **40**. Each auxiliary

panel **40** is, in this example, perforated to permit the free flow of air through the auxiliary panel **40**. In this manner, once installed in place, air may flow freely to and from within building via the perforated auxiliary panel **40** and past the sloped top and the soffit **90** itself. The sloped top is useful for preventing blockage of soffit **90** as compared to an unsloped top which would cover a segment of soffit **90** and therefore block the free flow of air. The sloped top also provides more insulation to the upper regions of the wall **120** of the building than would simply spacing a flat-topped insulating panel from the soffit **90**. This is because the sloped top enables the insulating panel P to reach to the top of the wall thereby providing some improved coverage of the wall than would the absence of insulation.

[0072] While system **100** is useful with one or more insulating panels P with sloped tops because airflow is unblocked while still providing at least some additional insulation to the top of the wall, alternatives are contemplated in which installations involve simply using auxiliary panel **40** with standard insulating panels (those having non-sloped tops), where the standard insulating panels are spaced from the soffit **90** to provide airflow that meets the applicable building code(s).

[0073] Furthermore, alternatives are contemplated in which just the sloped-top insulating panel P is employed or provided, without provision or installation of auxiliary panel **40**, in order to provide insulation without air blockage where the finishing provided by the auxiliary panel **40** is not necessarily required. Furthermore, a drainage or moisture management membrane may be placed over sheathing **122** to inhibit ingress of moisture into the sheathing **122**. The wall **120** supports rafters R which project outwardly to a fascia **124**.

[0074] As shown also in FIG. 5, system **100** includes bottom cap 5, which receives and thus “caps” the bottom of insulating panel P, while coaxing any moisture M that comes within the channel of bottom cap 5 frontwards to reach apertures **20** and thus be conveyed out of the channel and away from the wall **120** along a graded ground surface G.

[0075] System **100** enhances the insulation of the wall **120**, while providing—using bottom cap 5—both aesthetic and functional coverage/protection of the bottom of the insulating panel P that further enables moisture that may have found its way behind the coverage to escape from system **100**.

[0076] FIG. 6 is cross-sectional endwise elevation view of a bottom cap **50A** for a foam insulating panel, according to an alternative example. Bottom cap **50A** is very similar to bottom cap 5, in that it also includes a rear wall **12A**, a front wall **14A**, and a bottom wall **16A**, a set of pedestals **18A**, and a set of apertures **20A**, all in very similar relative position and orientation as those of bottom cap 5. However, bottom cap **50A** provides an obtuse internal angle **13A** that is less than obtuse internal angle **13**, and an acute internal angle **15A** that is accordingly greater than acute internal angle **15**. That is, the frontwards slope of bottom wall **16A** is less than that of bottom wall **16** of bottom cap 5. In this example, obtuse internal angle **13A** is 95 degrees and acute internal angle **15A** is 85 degrees. Because, in this example, the tops of both rear wall **12A** and front wall **14A** come to the same height when rear wall **12A** and front wall **14A** are both perfectly vertical, if rear wall **12A** is 1 inch tall, then with the 4 inch width W, front wall **14A** would be about 1.5 inches tall.

[0077] FIG. 7 is cross-sectional endwise elevation view of a bottom cap **50B** for a foam insulating panel, according to an alternative example. Bottom cap **50B** is very similar to bottom cap 5, in that it also includes a rear wall **12B**, a front wall **14B**, and a bottom wall **16B**, a set of pedestals **18B**, and a set of apertures **20B**, all in very similar relative position and orientation as those of bottom cap 5. However, bottom cap **50B** provides an obtuse internal angle **13B** that is less than obtuse internal angle **13**, and an acute internal angle **15B** that is accordingly greater than acute internal angle **15**. That is, the frontwards slope of bottom wall **16B** is less than that of bottom wall **16** of bottom cap 5. In this example, obtuse internal angle **13B** is 93 degrees and acute internal angle **15B** is 87 degrees. Because, in this example, the tops of both rear wall **12A** and front wall **14A** come to the same height when rear wall **12A** and front wall **14A** are both perfectly vertical, if rear wall **12A** is 1 inch tall, then with the 4 inch width W, front wall **14A** would be in the range of about 1 and

less than 1.5 inches tall.

[0078] FIG. **8** is cross-sectional endwise elevation view of a bottom cap **50C** for a foam insulating panel, according to an alternative example. Bottom cap **50C** is very similar to bottom cap **5**, in that it also includes a rear wall **12C**, a front wall **14C**, and a bottom wall **16C**, a set of pedestals **18C**, and a set of apertures **20C**, all in very similar relative position and orientation as those of bottom cap **5**. However, bottom cap **50C** provides an obtuse internal angle **13C** that is greater than obtuse internal angle **13**, and an acute internal angle **15C** that is accordingly less than acute internal angle **15**. That is, the frontwards slope of bottom wall **16C** is greater than that of bottom wall **16** of bottom cap **5**. In this example, obtuse internal angle **13C** is 100 degrees and acute internal angle **15C** is 80 degrees. Because, in this example, the tops of both rear wall **12C** and front wall **14C** come to the same height when rear wall **12C** and front wall **14C** are both perfectly vertical, if rear wall **12C** is 1 inch tall, then with the 4 inch width **W**, front wall **14C** would be greater than 1.85 inches tall.

[0079] FIG. **9** is cross-sectional endwise elevation view of a bottom cap **50D** for a foam insulating panel, according to an alternative example. Bottom cap **50D** is very similar to bottom cap **5**, in that it also includes a rear wall **12D**, a front wall **14D**, and a bottom wall **16D**, a set of pedestals **18D** (a “first” set of pedestals, in this example), and a set of apertures **20D**, all in very similar relative position and orientation as those of bottom cap **5**. Furthermore, bottom cap **50D** provides an obtuse internal angle **13C** that is the same as obtuse internal angle **13**, and an acute internal angle **15C** that is accordingly the same as acute internal angle **15**. That is, the frontwards slope of bottom wall **16D** is the same as that of bottom wall **16** of bottom cap **5**. However, in this example, bottom cap **50D** includes a second set of spaced pedestals **19D** extending into the open-topped channel from a region of bottom wall **16D** that is proximal to front wall **12D**. In this example, each of the spaced pedestals **19D** in the second set extends to a same height with respect to rear wall **12D** as the spaced pedestals **18D** in the first set. In particular, the bottom side of a foam insulating panel received within the open-topped channel of bottom cap **50D** would rest generally horizontally upon the first and second sets of pedestals **18D**, **19D**. As such, each of pedestals **19D** is itself taller than pedestals **18D**, but extends from a region of bottom wall **16D** that is lower in the slope.

[0080] FIG. **10** is a top plan view of bottom cap **50D**, showing bottom cap **50D** extending between a first end **8D** and a second end **10D**. As shown, the second set of pedestals **19D** extends along a side of the set of apertures **20D** that is opposite front wall **4D**. While alternatives to this configuration are possible, it is generally useful to have apertures **20D** as close to the lowest point of the sloped bottom wall **16D** as can be managed, subject to the strength and integrity of the bottom cap being maintained. Pedestals **19D** may be formed using a similar process to that of pedestals **18D** or may, due primarily to their different heights, be formed and/or arranged with respect to other features of bottom cap **50D** using a different process.

[0081] FIG. **11** is a front perspective view of bottom cap **50D** receiving a foam insulating panel **P** having a thickness of about 4 inches, corresponding to a width **W** of about 4 inches. It can be seen that, when seated within bottom cap **50D**, foam insulating panel **P** rests on pedestals **18D** (only one of which is visible in FIG. **11**) and also on pedestals **19D** (only one of which is visible in FIG. **11**), and closely between both rear wall **12D** and front wall **14D**, thus providing a generally wedge-shaped volume formed mainly by the bottom of foam insulating panel **P**, a portion of front wall **14D**, and bottom wall **16D**, in which moisture can flow generally freely underneath foam insulating panel **P** towards apertures **20D** (only one of which is visible in FIG. **11**) so that the moisture can escape to outside of the channel.

[0082] FIG. **12** is cross-sectional endwise elevation view of a bottom cap **50E** for a foam insulating panel, according to an alternative example. Bottom cap **50E** is very similar to bottom cap **5**, in that it also includes a rear wall **12E**, a front wall **14E**, and a bottom wall **16E**, and a set of apertures **20E**, all in very similar relative position and orientation as those of bottom cap **5**. In this example, however, rather than a set of spaced pedestals extending into the open-topped channel from a

region of the bottom wall **16E** that is proximal to the rear wall **12E**, a set of spaced ledges **21E** extends into the open-topped channel from a region of the elongate rear wall **12E** that is proximal to bottom wall **16E**. The spaced ledges **21E** together support a rear end of the bottom of a foam insulating panel when received within the open-topped channel so that the foam insulating panel does not rest directly on bottom wall **16E**. In this way, moisture that might come into the channel from the rear while the foam insulating panel is received within the channel is generally able to flow along bottom wall **16E**, underneath the foam insulating panel and through the spaces between ledges **21E**, towards front wall **14E**. Each one of ledges **21E** may be formed by a punch process that forms indentations into sheet metal from a rear-facing side of rear wall **12E**, thus forming ledges **21E** that extend laterally into the open-topped channel. Ledges **21E** may be formed by other processes.

[0083] Furthermore, bottom cap **50E** provides an obtuse internal angle **13E** that is the same as obtuse internal angle **13**, and an acute internal angle **15E** that is accordingly the same as acute internal angle **15**. That is, the frontwards slope of bottom wall **16E** is the same as that of bottom wall **16** of bottom cap **5**.

[0084] FIG. **13** is a top plan view of bottom cap **50E**, showing bottom cap **50E** extending between a first end **8E** and a second end **10E**. As shown, the second set of pedestals **19E** extends along a side of the set of apertures **20E** that is opposite front wall **14E**. While alternatives to this configuration are possible, it is generally useful to have apertures **20E** as close to the lowest point of the sloped bottom wall **16E** as can be managed, subject to the strength and integrity of the bottom cap being maintained.

[0085] FIG. **14** is cross-sectional endwise elevation view of a bottom cap **50F** for a foam insulating panel, according to an alternative example. Bottom cap **50F** is very similar to bottom cap **50E**, in that it also includes a rear wall **12E**, a front wall **14E**, and a bottom wall **16E**, a set of apertures **20E**, and a set of spaced ledges **21F** extending into the open-topped channel from a region of the elongate rear wall **12F** that is proximal to bottom wall **16F**, all in very similar relative position and orientation as those of bottom cap **50E**. However, in this example, like bottom cap **50D**, bottom cap **50F** includes a set of spaced pedestals **19F** extending into the open-topped channel from a region of bottom wall **16F** that is proximal to front wall **12F**. In this example, each of the spaced pedestals **19F** in the set extends to a same height with respect to rear wall **12F** as the tops of spaced ledges **21F**. In particular, the bottom side of a foam insulating panel received within the open-topped channel of bottom cap **50F** would rest generally horizontally upon the set of ledges **21F** and the set of pedestals **19F**.

[0086] The spaced ledges **21F** and the pedestals **19F** together support the bottom of a foam insulating panel when received within the open-topped channel so that the foam insulating panel does not rest directly on bottom wall **16F**. In this way, moisture that might come into the channel from the rear while the foam insulating panel is received within the channel is generally able to flow along in inside facing surface of rear wall **12F**, between ledges **21E**, and along bottom wall **16F** underneath the foam insulating panel and to reach the apertures **20F**.

[0087] Furthermore, bottom cap **50E** provides an obtuse internal angle **13E** that is the same as obtuse internal angle **13**, and an acute internal angle **15E** that is accordingly the same as acute internal angle **15**. That is, the frontwards slope of bottom wall **16E** is the same as that of bottom wall **16** of bottom cap **5**.

[0088] FIG. **15** is a top plan view of bottom cap **50E**, showing bottom cap **50E** extending between a first end **8E** and a second end **10E**. As shown, the second set of pedestals **19E** extends along a side of the set of apertures **20E** that is opposite front wall **14E**. While alternatives to this configuration are possible, it is generally useful to have apertures **20E** as close to the lowest point of the sloped bottom wall **16E** as can be managed, subject to the strength and integrity of the bottom cap being maintained.

[0089] Although examples have been described, those of skill in the art will appreciate that

variations and modifications may be made without departing from the spirit, scope and purpose of the disclosure as defined by the appended claims.

[0090] For example, it will be appreciated that, in examples, the tops of front and rear walls do not necessarily need to reach the same heights when the front and rear walls are perfectly vertical. For example, a rear wall in an example may rise to a lesser height than the front wall, or may rise to a greater height than the front wall. However, it will be appreciated that there should be provided a region of the rear wall that sufficiently contains the rear of a foam insulating panel when the bottom of the panel—typically a horizontal or near-horizontal bottom—is seated inside the bottom cap while a region of the front wall sufficiently contains the front of the same foam insulating panel.

[0091] FIG. **16** is cross-sectional endwise elevation view of a bottom cap **50G** for a foam insulating panel, according to an alternative example. Bottom cap **50G** is very similar to bottom cap **5**, in that it also includes a rear wall **12G**, a front wall **14G**, and a bottom wall **16G**, a set of pedestals **18G**, and a set of apertures **20G**, all in very similar relative position and orientation as those of bottom cap **5**. Similarly, bottom cap **50G** provides an obtuse internal angle that is the same as obtuse internal angle **13**, and an acute internal angle that is accordingly the same as acute internal angle **15**. However, in this example, the tops of rear wall **12G** and front wall **14G** come to different heights when rear wall **12G** and front wall **14G** are both perfectly vertical. In particular, front wall **14G** comes to a greater height than does rear wall **12G**.

[0092] FIG. **17** is cross-sectional endwise elevation view of a bottom cap **50H** for a foam insulating panel, according to an alternative example. Bottom cap **50H** is very similar to bottom cap **5**, in that it also includes a rear wall **12H**, a front wall **14H**, and a bottom wall **16H**, a set of pedestals **18H**, and a set of apertures **20H**, all in very similar relative position and orientation as those of bottom cap **5**. Similarly, bottom cap **50H** provides an obtuse internal angle that the same as obtuse internal angle **13**, and an acute internal angle that is accordingly the same as acute internal angle **15**. However, in this example, the tops of rear wall **12H** and front wall **14H** come to different heights when rear wall **12H** and front wall **14H** are both perfectly vertical. In particular, front wall **14H** comes to a lesser height than does rear wall **12H**.

[0093] FIG. **18** is cross-sectional endwise elevation view of a bottom cap **50I** for a foam insulating panel, according to an alternative example. Bottom cap **50I** is very similar to bottom cap **5**, in that it also includes a rear wall **12I**, a front wall **14I**, and a bottom wall **16I**, a set of pedestals **18I**, and a set of apertures **20I**, all in very similar relative position and orientation as those of bottom cap **5**. Similarly, bottom cap **50I** provides an obtuse internal angle that the same as obtuse internal angle **13**, and an acute internal angle that is accordingly the same as acute internal angle **15**. However, in this example, the tops of rear wall **12I** and front wall **14I** come to different heights when rear wall **12I** and front wall **14I** are both perfectly vertical. In particular, front wall **14I** comes to a lesser height than does rear wall **12I**. Furthermore, each of rear wall **12I** and front wall **14I** have extensions down from bottom wall **16I**, which may be useful functionally or aesthetically.

[0094] FIG. **19** is cross-sectional endwise elevation view of a bottom cap **50J** for a foam insulating panel, according to an alternative example. Bottom cap **50J** is very similar to bottom cap **5**, in that it also includes a rear wall **12J**, a front wall **14J**, and a bottom wall **16J**, a set of pedestals **18J**, and a set of apertures **20J**, all in very similar relative position and orientation as those of bottom cap **5**. Similarly, bottom cap **50J** provides an obtuse internal angle that the same as obtuse internal angle **13**, and an acute internal angle that is accordingly the same as acute internal angle **15**. However, in this example, the tops of rear wall **12J** and front wall **14J** come to different heights when rear wall **12J** and front wall **14J** are both perfectly vertical. In particular, front wall **14J** comes to a lesser height than does rear wall **12J**. Furthermore, as compared with bottom cap **50I** of FIG. **18**, only front wall **14J** has an extension down from bottom wall **16J**, which may be useful functionally or aesthetically.

[0095] Various further combinations of aspects disclosed herein are possible as examples, such as

the examples set forth in the combinations of multiple clauses below.

Clauses

[0096] Clause 1. A bottom cap for a foam building panel, the bottom cap comprising:

[0097] an elongate rear wall; [0098] an elongate front wall spaced from and extending parallel to the elongate rear [0099] a bottom wall extending at an obtuse inside angle from the elongate rear wall to the elongate front wall thereby to form an open-topped channel; [0100] a first set of spaced pedestals extending into the open-topped channel from a region of the bottom wall that is proximal to the elongate rear wall; and [0101] a set of spaced apertures extending from inside the open-topped channel through a region of the bottom wall that is proximal to the elongate front wall.

[0102] Clause 2. The bottom cap of clause 1, wherein the obtuse inside angle is from about 95 degrees to about 105 degrees.

[0103] Clause 3. The bottom cap of clause 2, wherein the obtuse inside angle is from about 97 degrees to about 102 degrees.

[0104] Clause 4. The bottom cap of clause 3, wherein the obtuse inside angle is 97 degrees.

[0105] Clause 5. The bottom cap of clause 3, wherein the obtuse inside angle is 102 degrees.

[0106] Clause 6. The bottom cap of clause 1, further comprising: [0107] a second set of spaced pedestals extending into the open-topped channel from a region of the bottom wall that is proximal to the elongate front wall, wherein each of the spaced pedestals in the second set extends to a same height with respect to the elongate rear wall as the spaced pedestals in the first set.

[0108] Clause 7. The bottom cap of clause 1, wherein the elongate front wall is spaced from the elongate rear wall by about 4 inches.

[0109] Clause 8. The bottom cap of clause 7, wherein the elongate rear wall extends to about 1 inch from the bottom wall.

[0110] Clause 9. The bottom cap of clause 8, wherein the elongate front wall extends to about 1.5 inches from the bottom wall.

[0111] Clause 10. The bottom cap of clause 8, wherein the elongate front wall extends to about 1.85 inches from the bottom wall.

[0112] Clause 11. A bottom cap for a foam building panel, the bottom cap comprising: [0113] an elongate rear wall; [0114] an elongate front wall spaced from and extending parallel to the elongate rear [0115] a bottom wall extending at an obtuse inside angle from the elongate rear wall to the elongate front wall thereby to form an open-topped channel; [0116] a set of spaced ledges extending into the open-topped channel from the elongate rear wall; and [0117] a set of spaced apertures extending from inside the open-topped channel through a region of the bottom wall that is proximal to the elongate front wall.

[0118] Clause 12. The bottom cap of clause 11, wherein the obtuse inside angle is from about 95 degrees to about 105 degrees.

[0119] Clause 13. The bottom cap of clause 12, wherein the obtuse inside angle is from about 97 degrees to about 102 degrees.

[0120] Clause 14. The bottom cap of clause 13, wherein the obtuse inside angle is 97 degrees.

[0121] Clause 15. The bottom cap of clause 13, wherein the obtuse inside angle is 102 degrees.

[0122] Clause 16. The bottom cap of clause 11, further comprising: [0123] a set of spaced pedestals extending into the open-topped channel from a region of the bottom wall that is proximal to the elongate front wall, wherein each of the spaced pedestals in the second set extends to a same height with respect to the elongate rear wall as the spaced ledges.

[0124] Clause 17. The bottom cap of clause 11, wherein the elongate front wall is spaced from the elongate rear wall by about 4 inches.

[0125] Clause 18. The bottom cap of clause 17, wherein the elongate rear wall extends to about 1 inch from the bottom wall.

[0126] Clause 19. The bottom cap of clause 18, wherein the elongate front wall extends to about 1.5 inches from the bottom wall.

[0127] Clause 20. The bottom cap of clause 18, wherein the elongate front wall extends to about 1.85 inches from the bottom wall.

Claims

1. A bottom cap for a foam building panel, the bottom cap comprising: an elongate rear wall; an elongate front wall spaced from and extending parallel to the elongate rear wall; a bottom wall extending at an obtuse inside angle from the elongate rear wall to the elongate front wall thereby to form an open-topped channel; a first set of spaced pedestals extending into the open-topped channel from a region of the bottom wall that is proximal to the elongate rear wall; and a set of spaced apertures extending from inside the open-topped channel through a region of the bottom wall that is proximal to the elongate front wall.
2. The bottom cap of claim 1, wherein the obtuse inside angle is from about 95 degrees to about 105 degrees.
3. The bottom cap of claim 2, wherein the obtuse inside angle is from about 97 degrees to about 102 degrees.
4. The bottom cap of claim 3, wherein the obtuse inside angle is 97 degrees.
5. The bottom cap of claim 3, wherein the obtuse inside angle is 102 degrees.
6. The bottom cap of claim 1, further comprising: a second set of spaced pedestals extending into the open-topped channel from a region of the bottom wall that is proximal to the elongate front wall, wherein each of the spaced pedestals in the second set extends to a same height with respect to the elongate rear wall as the spaced pedestals in the first set.
7. The bottom cap of claim 1, wherein the elongate front wall is spaced from the elongate rear wall by about 4 inches.
8. The bottom cap of claim 7, wherein the elongate rear wall extends to about 1 inch from the bottom wall.
9. The bottom cap of claim 8, wherein the elongate front wall extends to about 1.5 inches from the bottom wall.
10. The bottom cap of claim 8, wherein the elongate front wall extends to about 1.85 inches from the bottom wall.
11. A bottom cap for a foam building panel, the bottom cap comprising: an elongate rear wall; an elongate front wall spaced from and extending parallel to the elongate rear wall; a bottom wall extending at an obtuse inside angle from the elongate rear wall to the elongate front wall thereby to form an open-topped channel; a set of spaced ledges extending into the open-topped channel from the elongate rear wall; and a set of spaced apertures extending from inside the open-topped channel through a region of the bottom wall that is proximal to the elongate front wall.
12. The bottom cap of claim 11, wherein the obtuse inside angle is from about 95 degrees to about 105 degrees.
13. The bottom cap of claim 12, wherein the obtuse inside angle is from about 97 degrees to about 102 degrees.
14. The bottom cap of claim 13, wherein the obtuse inside angle is 97 degrees.
15. The bottom cap of claim 13, wherein the obtuse inside angle is 102 degrees.
16. The bottom cap of claim 11, further comprising: a set of spaced pedestals extending into the open-topped channel from a region of the bottom wall that is proximal to the elongate front wall, wherein each of the spaced pedestals in the set of spaced pedestals extends to a same height with respect to the elongate rear wall as the spaced ledges.
17. The bottom cap of claim 11, wherein the elongate front wall is spaced from the elongate rear wall by about 4 inches.
18. The bottom cap of claim 17, wherein the elongate rear wall extends to about 1 inch from the bottom wall.

19. The bottom cap of claim 18, wherein the elongate front wall extends to about 1.5 inches from the bottom wall.

20. The bottom cap of claim 18, wherein the elongate front wall extends to about 1.85 inches from the bottom wall.
